

Progress Report

Smart Boxing Glove

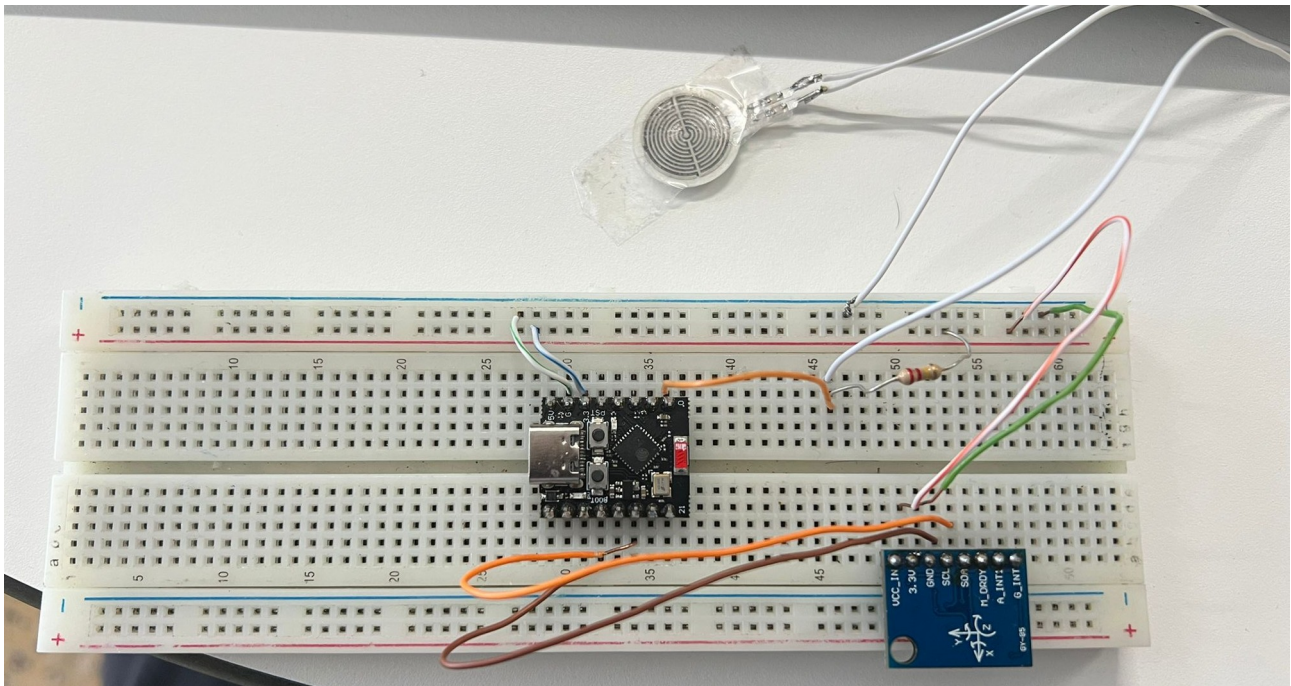
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1 Introduction

1.1 Background and Motivation

In recent years, technology has become increasingly important in sports training, enabling more precise and data driven performance analysis. Although sports like football already make extensive use of these tools, boxing still relies largely on visual assessment by coaches, lacking accurate measurement of key factors such as punch speed, acceleration, and impact force.

This project is motivated by the growing interest in accessible, data based training solutions. Existing technologies are often expensive or limited, so the goal is to develop a flexible and affordable system that can be used in different training environments. Additionally, our interest in boxing and the desire to support athletes who train independently inspired us to create a solution that helps improve performance while applying the technical skills we have developed.

1.2 Objectives

The main objective of this project is to design and develop a smart boxing glove capable of acquiring, transmitting, and analyzing motion and force data in real time. The system aims to provide an objective and data driven approach to boxing training, enabling improved performance evaluation and technique analysis.

To achieve this goal, the following specific objectives are defined:

- Develop an embedded system capable of acquiring data from multiple sensors, including an accelerometer, a gyroscope, and a force sensor
- Implement a reliable data acquisition strategy with multiple sampling rates to accurately capture the dynamics of a boxing punch
- Optimize the communication process to handle throughput and latency constraints inherent to wireless systems
- Develop a software architecture based on a state driven approach to ensure robustness and proper handling of system events
- Enable data visualization and interpretation to support performance analysis and training feedback

Additionally, the project aims to explore trade offs between sampling frequency, data size, transmission reliability, and system latency, ensuring that the final solution remains both efficient and scalable.

1.3 Document Structure

This document is organized into several sections, each addressing a specific aspect of the development progress:

Section 1 introduces the context, motivation, and objectives of the project.

Section 2 provides a high-level System Overview, identifying the hardware components and the general data flow from the glove to the user.

Section 3 details the Software Architecture, covering connectivity management, data processing pipelines, and the 3D visualization logic.

Section 4 describes the Hardware Architecture, focusing on peripheral remapping, the sensor synchronization strategy, and the power subsystem.

Section 5 presents the Implementation and Results, detailing the development of I2C drivers for each sensor and showing the current system output.

Section 6 summarizes the Tasks Completed, including battery sizing, strike duration analysis, FSR characterization, and the application preview.

Section 7 identifies the Technical Challenges and problems encountered during development, such as the gyroscope data stagnation.

Section 8 outlines the Project Management, including the work distribution between members and the planning for upcoming phases.

Section 9 concludes the report with final remarks and provides the link to the project's source code.

2 System Overview

The proposed system is an integrated wearable solution designed to monitor and quantify boxing performance through high-speed data acquisition and wireless transmission. The architecture is divided into three main layers: the sensing layer, the processing and control layer, and the visualization layer. From now on the system will be called "Smart Glove"

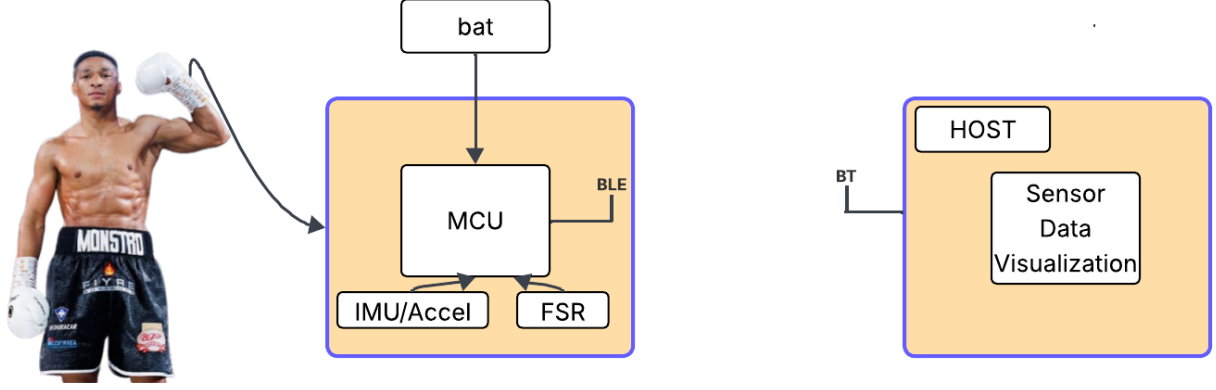


Figure 1: High Architecture System

2.1 Hardware Components

This section summarizes the hardware selected for the prototype. The focus was on small dimensions for glove integration and efficient wireless communication.

- **Microcontroller (ESP32-C3 SuperMini):** A high-performance 32-bit RISC-V core with integrated Bluetooth 5.0 (BLE). Selected for its ultra-small footprint and low power consumption.
- **IMU Module (GY-85):** A 9-DOF unit combining three distinct digital sensors via a shared I2C bus:
 - **ADXL345:** Linear acceleration and impact detection.
 - **ITG3200:** Angular velocity for punch rotation tracking.
 - **HMC5883L:** Magnetic field sensing for orientation stability.
- **Impact Sensor (FSR 402):** A thin, flexible piezoresistive analog sensor used to measure the force of the strike.

Table 1: Hardware Specifications and Project Function

Component	Main Feature	Function
ESP32-C3	Bluetooth 5.0 (BLE)	Central processing and wireless link.
ADXL345	10-bit Resolution	Captures punch acceleration and impact.
ITG3200	$\pm 2000^\circ/s$ Range	Tracks high-speed hand rotations.
HMC5883L	3-Axis Digital Compass	Provides absolute orientation reference.
FSR 402	Analog Resistance	Measures impact intensity and peak force.

2.2 General View

The Smart Glove is designed to provide real-time feedback on boxing performance by capturing, processing, and visualizing data from each punch. The process follows a clear path, from physical movement to the final digital display.

First, the Sensing and Processing Phase occurs in the glove. A set of sensors captures the action: Force Sensing Resistors (FSRs) measure the intensity of the impact, while an Inertial Measurement Unit (IMU) tracks movement and orientation. The ESP32-C3 microcontroller manages the configuration of these sensors and handles the raw data.

Next, the Data Transmission Phase uses the NimBLE stack to establish a Bluetooth Low Energy (BLE) connection. The microcontroller wirelessly sends the processed metrics and trajectory data to a receiving device (smartphone or PC), ensuring that the athlete's mobility is not limited by cables.

Finally, the Visualization Phase occurs on the receiver side. The software decodes the received data to display the final results:

Force: The impact value already converted to kilograms (kg).

Velocity: The maximum speed reached during the movement.

3D Trajectory: A graphical representation of the strike trajectory in three planes (X, Y, Z), showing the movement from the guard position to the target.

3 Software Architecture

3.1 Connectivity and Device Management

The application is responsible for discovering, selecting, and maintaining communication with Bluetooth Low Energy devices corresponding to the glove system. The process begins with continuous scanning of nearby devices, followed by a filtering stage that identifies compatible devices based on predefined characteristics such as name, service identifiers, or signal strength.

If no compatible device is found, the application remains in scanning mode until a valid device is detected. Once available devices are identified, they are presented to the user, who can select the desired device for connection.

After selection, the application attempts to establish a connection with the device. In case of failure, an error is reported and the system returns to the selection phase. When the connection is successfully established, a validation step is performed through the exchange of test data to ensure reliable communication.

Adaptive Communication The application dynamically adjusts its communication parameters based on the number of connected devices and current system conditions, optimizing performance and resource utilization.

Single Device Mode When a single device is connected, the application enables a high performance mode. In this configuration, higher data rates and larger transmission payloads are used, maximizing the available bandwidth and improving data throughput.

Multi-Device Mode When multiple devices are connected, the application switches to a balanced mode of operation. In this case, communication parameters are adjusted to ensure fair bandwidth distribution, preventing congestion and maintaining stable performance across all devices.

3.2 Data Processing and Validation Pipeline

Once the connection is established and validated, the application initiates the data acquisition phase. The device is configured with the required operating parameters, including sampling rate and data transmission settings, and the system begins receiving data streams.

Initial Configuration Before entering continuous operation, the application sends configuration commands to the device, defining parameters such as sampling frequency and data packet structure. This ensures that the incoming data matches the expected format and timing constraints.

Integrity Check Each received data packet undergoes a validation process to ensure data integrity and consistency. This includes verification of format correctness and error detection mechanisms. If a packet is identified as invalid or corrupted, it is discarded, preventing incorrect data from propagating through the system.

Valid data is then decompressed and converted into structured internal representations suitable for further processing.

Processing Pipeline After validation, the data is processed to extract meaningful information. This includes the computation of metrics, detection of relevant events such as punch movements, and preparation of data for visualization. The processing stage ensures that raw sensor data is transformed into a usable and interpretable form.

3.3 Visualization and 3D Reconstruction

The application includes a visualization subsystem responsible for representing the motion of the glove in three dimensions. After the initial configuration phase, the graphical environment is initialized, including the setup of rendering components and scene structures.

Based on the processed data, the application reconstructs the movement of the glove in real time. This allows the trajectory of each punch to be visualized dynamically, providing immediate feedback to the user.

The visualization is continuously updated as new data is received, ensuring a smooth and accurate representation of motion. Additionally, the system supports real time parameter adjustments, allowing the user or the application itself to modify operational settings without interrupting the data stream.

3.4 Application

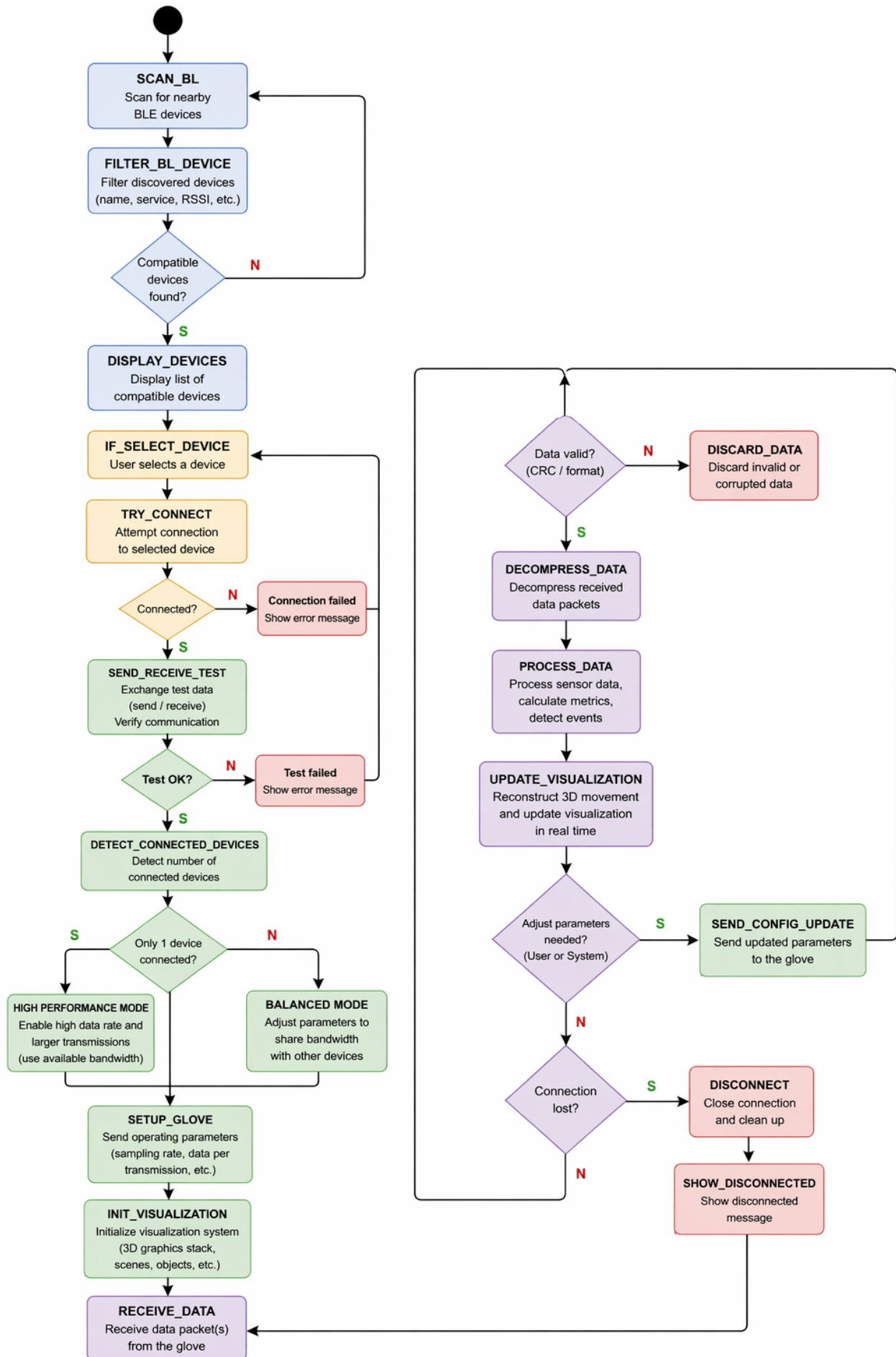


Figure 2: Software Diagram

The application begins by scanning for Bluetooth Low Energy devices, filtering the available devices in order to identify those corresponding to the glove. If no compatible device is found, the application remains in scanning mode until a valid device is detected.

Once a device is detected, the application establishes a connection with the glove and starts receiving data. During this phase, an initial configuration process is performed, where the application sends the operating parameters to the device, such as the sampling rate and the amount of data per transmission.

If only one device is connected, the application activates a more efficient operating mode, allowing higher data rates and larger data transmissions, taking advantage of the available bandwidth. If multiple devices are connected, the application adjusts the parameters to ensure a balanced distribution of communication resources.

After the connection is established, the application initializes the visualization system, including the graphical stack responsible for the three dimensional representation of the movement. The application then enters an active waiting state, continuously receiving data from the glove.

The received data is processed, validated, and organized into internal structures. If invalid or corrupted data is detected, it is discarded, ensuring that only consistent information is used for processing and visualization.

Based on the valid data, the application reconstructs the movement in three dimensions, allowing real time visualization of the punch trajectory.

During execution, the application allows dynamic adjustment of operating parameters, sending updated values to the device as needed. These adjustments can be performed by the user or automatically, based on system performance conditions.

4 Hardware Architecture

The hardware architecture of this prototype was designed to ensure high fidelity data acquisition and minimal space occupation. The Smart Glove is based on an ESP32-C3 SuperMini microcontroller, which interacts with a set of sensors through analog and digital protocols.

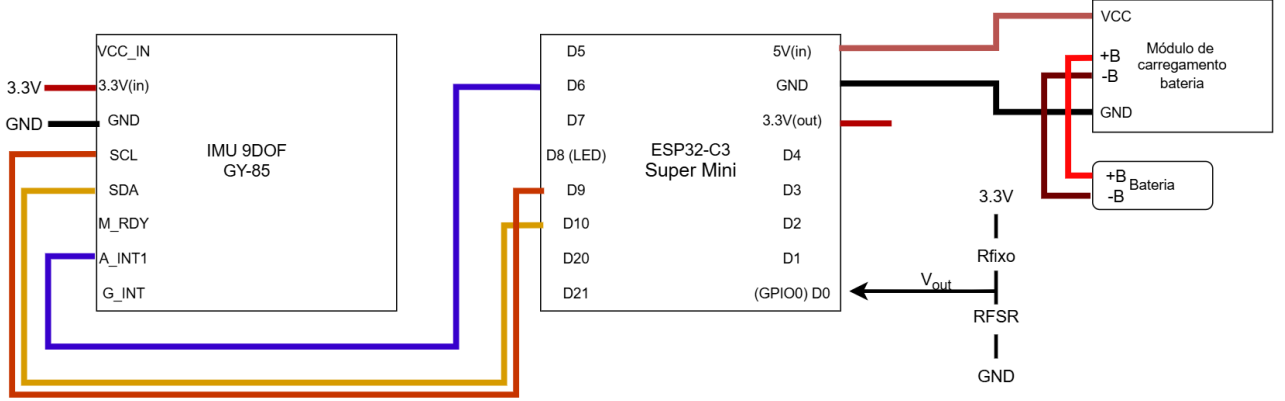


Figure 3: HW connections and architecture

4.1 Peripheral Remapping

To avoid hardware conflicts, the standard I2C pinout was remapped. The default SDA (GPIO8) was avoided as it is tied to an internal LED on the SuperMini board, by mapping the SDA line to GPIO10, the internal LED remains available for the still not implemented functions.

4.2 Sensor Synchronization Strategy

The Smart Glove architecture was designed to leverage a hardware synchronization strategy, ensuring temporal alignment between all sensing modules. The ADXL345 accelerometer was selected as the system's clock because its sampling frequency is lower than that of the gyroscope.

Using the hardware 'Data Ready' interrupt (A_INT1), the firmware aims to trigger the simultaneous capture of the 3 IMU sensors. Relying on hardware timed signals instead of software based polling ensures that the data acquisition process operates at the sensors' peak physical output rate.

4.3 Power Subsystem

The architecture includes a dedicated power path designed for portability and field use. The Smart Glove is intended to be powered by a Lithium-Polymer (LiPo) battery.

5 Implementation and Results

In this project phase, we focused on the implementation and integration of hardware and software for data acquisition and communication, ensuring that the sensors can capture the user’s movements and impacts. The main actions and resolutions are detailed in the following points.

5.1 I2C Communication

A dedicated I2C driver has been implemented to enable communication between the ESP32-C3 and all digital sensors. The bus operates in master mode, with the SDA and SCL lines configured using the predefined pin mapping, ensuring compatibility with the board.

To simplify interaction with different sensors, generic read and write functions have been implemented. These functions encapsulate data transmission and reception, allowing register-based communication via simple function calls.

The driver supports multi-byte read and single-byte write operations, which is sufficient for all connected peripherals. This abstraction allows higher-level modules, such as the accelerometer and magnetometer drivers, to access sensor data without directly handling low-level I2C operations.

5.2 FSR sensor (Impact measurement)

We configured the microcontroller’s Analog-to-Digital Converter (ADC) to read the Force Sensing Resistor (FSR). The channel attenuation was set to 12dB (allowing voltage measurements up to 3.3V), with a 12-bit resolution, and we also calculated the optimal pull-up resistor value for the voltage divider using the geometric mean formula ($R_{pullup} = \sqrt{R_{max} \times R_{min}}$).

The microcontroller correctly reads the force applied to the sensor on the ADC scale (0 to 4095). Since a voltage divider with a pull-up resistor was implemented, the raw ADC reading is inversely proportional to the applied force. To correct for this, the software logic was adjusted to invert the scale, ensuring that the output reflects the actual impact, higher values for stronger punches.

5.3 Accelerometer (ADXL345)

A dedicated driver for the accelerometer was implemented that utilizes the established I2C bus. During initialization, the sensor was configured with a sampling rate of 100 Hz, its resolution scale was set to $\pm 2g$ (for initial baseline testing). After the configuration phase, the device was successfully transitioned into active measurement mode.

The Smart Glove successfully reads the raw data across all three axes (X, Y, and Z) and applies the appropriate mathematical scale factor to convert these raw readings into real acceleration values in units of *g-force*.

5.4 Gyroscope (ITG3200)

The ITG3200 was integrated into the I2C bus to track the angular velocity of the glove. The initialization sequence followed the manufacturer’s requirements: a hardware reset, a stabilization delay, and the selection of the X-axis gyroscope as the PLL clock reference. The sensor was configured for a 100 Hz sampling rate with an internal digital low-pass filter.

Initial tests confirmed that the physical wiring and I2C addressing are correct, as the microcontroller successfully retrieved the Device ID (0x68) via a "Who Am I" register check.

5.5 Magnetometer (HMC5883L)

To follow orientation, the HMC5883L was added to the I2C. The initialization sequence configures the sensor for 8-sample averaging at a rate of 15Hz and settles to continuous measurement mode.

During data extraction, the parsing logic was specifically written to handle the sensor’s unconventional data output order (X, Z, Y), each byte is correctly mapped to its respective axis variable. Finally, the raw values are divided by the configured gain of 1090 LSb/Gauss to output the magnetic field strength in Gauss.

5.6 Wireless Communication (NimBLE)

Wireless communication was established using the NimBLE stack, selected for its efficient memory management on the ESP32-C3. The implementation follows a strict initialization sequence, starting with the activation of the non-volatile storage module (NVS), a mandatory requirement for the storage configuration stack and pairing keys.

The driver configures the basic GAP (Generic Access Profile) and GATT (Generic Attribute Profile) services, setting the system identifier as "LUVA BOX 2026". To ensure that data transmission does not block critical sensor acquisition tasks, the stack is implemented in a dedicated FreeRTOS task, initiated after successful Bluetooth host configuration.

Connectivity logic is based on a callback mechanism for event management. After initial synchronization (on_sync), the device enters a non-targeted advertising state, allowing its discovery by external applications. To ensure system robustness during training, the innovative state machine in gap_event_cb detects connection failures or disconnections, automatically restarting the advertising process to keep the system available for reconnection without manual intervention.

5.7 System Integration and Final Data Output

The following figure illustrates the current real-time data output from the system. Demonstrates that the communication bus is fully operational, successfully capturing data from the FSR (Impact), ADXL345 (Acceleration), and HMC5883L (Magnetic Field).

```
FSR: 0 | ACC [X: -0.08 Y: -0.18 Z: -0.66] | GIRO [X: 0.00 Y: 0.00 Z: 0.00] | MAG [X: -0.25 Y: -0.51 Z: -0.44]
FSR: 0 | ACC [X: 1.61 Y: 0.56 Z: -1.01] | GIRO [X: 0.00 Y: 0.00 Z: 0.00] | MAG [X: -0.34 Y: -0.41 Z: -0.54]
FSR: 0 | ACC [X: -0.18 Y: 1.38 Z: -1.86] | GIRO [X: 0.00 Y: 0.00 Z: 0.00] | MAG [X: -0.33 Y: -0.39 Z: -0.46]
FSR: 0 | ACC [X: -0.67 Y: 1.15 Z: -1.16] | GIRO [X: 0.00 Y: 0.00 Z: 0.00] | MAG [X: -0.34 Y: -0.31 Z: -0.45]
FSR: 0 | ACC [X: -2.00 Y: 1.37 Z: 0.01] | GIRO [X: 0.00 Y: 0.00 Z: 0.00] | MAG [X: -0.31 Y: -0.31 Z: -0.43]
FSR: 0 | ACC [X: -2.00 Y: 0.91 Z: -1.15] | GIRO [X: 0.00 Y: 0.00 Z: 0.00] | MAG [X: -0.35 Y: -0.39 Z: -0.30]
FSR: 0 | ACC [X: -0.71 Y: -0.68 Z: 0.29] | GIRO [X: 0.00 Y: 0.00 Z: 0.00] | MAG [X: -0.29 Y: -0.37 Z: -0.42]
FSR: 0 | ACC [X: 0.49 Y: -0.54 Z: -0.71] | GIRO [X: 0.00 Y: 0.00 Z: 0.00] | MAG [X: -0.27 Y: -0.45 Z: -0.47]
FSR: 0 | ACC [X: 0.14 Y: -0.21 Z: -0.46] | GIRO [X: 0.00 Y: 0.00 Z: 0.00] | MAG [X: -0.27 Y: -0.46 Z: -0.46]
FSR: 0 | ACC [X: -0.98 Y: 0.46 Z: -0.44] | GIRO [X: 0.00 Y: 0.00 Z: 0.00] | MAG [X: -0.39 Y: -0.33 Z: -0.55]
FSR: 0 | ACC [X: 1.41 Y: 0.36 Z: -0.87] | GIRO [X: 0.00 Y: 0.00 Z: 0.00] | MAG [X: -0.42 Y: -0.64 Z: -1.16]
FSR: 0 | ACC [X: -0.93 Y: 0.06 Z: -0.47] | GIRO [X: 0.00 Y: 0.00 Z: 0.00] | MAG [X: -0.03 Y: -0.37 Z: -0.33]
FSR: 0 | ACC [X: 0.85 Y: -0.06 Z: 0.68] | GIRO [X: 0.00 Y: 0.00 Z: 0.00] | MAG [X: -0.18 Y: -0.50 Z: -1.50]
FSR: 0 | ACC [X: -0.87 Y: -0.28 Z: -0.48] | GIRO [X: 0.00 Y: 0.00 Z: 0.00] | MAG [X: -0.18 Y: -0.34 Z: -0.44]
FSR: 0 | ACC [X: -1.42 Y: -0.14 Z: -1.47] | GIRO [X: 0.00 Y: 0.00 Z: 0.00] | MAG [X: -0.22 Y: -0.33 Z: -0.41]
FSR: 0 | ACC [X: 0.06 Y: -0.79 Z: 0.77] | GIRO [X: 0.00 Y: 0.00 Z: 0.00] | MAG [X: -0.32 Y: -0.19 Z: -1.22]
FSR: 0 | ACC [X: 0.19 Y: -0.20 Z: -1.45] | GIRO [X: 0.00 Y: 0.00 Z: 0.00] | MAG [X: -0.43 Y: -0.58 Z: -0.40]
FSR: 0 | ACC [X: 1.00 Y: 0.13 Z: 0.75] | GIRO [X: 0.00 Y: 0.00 Z: 0.00] | MAG [X: -0.35 Y: -0.62 Z: -1.26]
FSR: 0 | ACC [X: -0.04 Y: 0.91 Z: -1.68] | GIRO [X: 0.00 Y: 0.00 Z: 0.00] | MAG [X: -0.36 Y: -0.57 Z: -0.47]
FSR: 0 | ACC [X: -0.15 Y: -0.12 Z: -1.18] | GIRO [X: 0.00 Y: 0.00 Z: 0.00] | MAG [X: -0.26 Y: -0.60 Z: -0.48]
FSR:4067 | ACC [X: 0.05 Y: -0.15 Z: -0.83] | GIRO [X: 0.00 Y: 0.00 Z: 0.00] | MAG [X: -0.34 Y: -0.50 Z: -0.53]
GOLPE DETETADO! FSR: 4067
```

Figure 4: System output showing active data from FSR, Accelerometer, and Magnetometer (Gyroscope values currently pending calibration).

6 Tasks Completed

To bring the project closer to a real-world use case, several measurements and tests were conducted during the hardware characterization and power consumption analysis phase.

- **Battery and Power Management:** To implement a fully portable system for athlete training, mobility is a critical requirement, which excludes the use of wired power supplies.

To properly size the required battery, we conducted a total current consumption analysis, for the sensors, the consumption values were sourced directly from their respective datasheets. Regarding the ESP32-C3 microcontroller, following our advisor's guidance to simulate a maximum power consumption scenario, we implemented a test application running a continuous Wi-Fi scanning loop, using a USB Tester, the average system consumption was measured. The battery was sized to ensure a 2-hour training duration, including a 50% safety margin, resulting in a required capacity of 231 mAh.

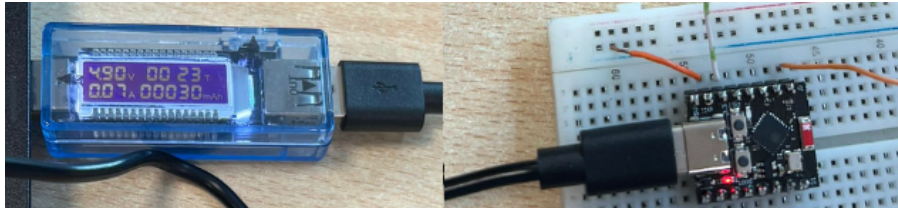


Figure 5: USB-Tester measuring ESP32-C3 mAh

- **Strike Execution Analysis:** To determine the required sampling rate for the sensors and the total duration of the longest maneuvers, a test was conducted with athletes from a local combat sports academy. High-frame-rate slow-motion video was used to precisely measure the time interval from the moment the strike leaves the guard position to the point of impact on the heavy bag. This data is essential for sizing the data buffer required for future trajectory reconstruction and analysis of the motion.

Table 2: Strike Analysis Results, Fastest and Average

Nome	Tipo de golpe	1ª Repetição	2ª Repetição	3ª Repetição	4ª Repetição	5ª Repetição	6ª Repetição	7ª Repetição	8ª Repetição	9ª Repetição	10ª Repetição	Média (segundos)	Categoria
Atleta 1	Jab	0,35	0,22	0,26	0,3	0,26	0,21	0,2	0,18	0,2	0,34	0,252	Manutenção
Atleta 1	Direto	0,4	0,25	0,3	0,27	0,32	0,33	0,35	0,31	0,35	0,27	0,315	Manutenção
Atleta 1	Cross	0,4	0,36	0,4	0,36	0,32	0,34	0,27	0,3	0,26	0,36	0,337	Manutenção
Atleta 1	Uppercut	0,36	0,41	0,33	0,34	0,37	0,34	0,31	0,32	0,38	0,3	0,346	Manutenção

Média de tempo de golpe				Golpe mais rápido		
Tipo de golpe	Competição	Manutenção	Média (segundos)	Tipo de golpe	Nome	Média (segundos)
Jab	0,275	0,268	0,271	**Jab**	Atleta 4	0,232
Direto	0,258	0,304	0,281	**Direto**	Atleta 6	0,238
Cross	0,383	0,446	0,415	**Cross**	Atleta 1	0,337
Uppercut	0,356	0,419	0,387	**Uppercut**	Atleta 2	0,293

Figure 6: Average Strike Time And Fastest Attack

- **Characterization of the FSR sensor:** To translate the raw ADC values into standard mass units (kg), a characterization process was performed using a digital scale as reference. The ADC readings were mapped against known weights to identify the sensor's sensitivity and saturation points. By applying a linear regression to the cleaned dataset (post IQR filtering), a mathematical transfer function was derived, allowing unit conversion within the firmware.

Table 3: Mapping of Known Weights to Raw ADC Values

Peso (Kg)	Valor ADC		
	Sensor 6Kg (ADC)	Sensor 15Kg (ADC)	Sensor Fita (ADC)
0	0	0	0
0,5	567	-	-
1	1715,29	-	-
2	2398,8	431,27	2240,86
3	3289	1157,33	2811,75
4	3496,31	1934,25	3557,92
5	3651	2147,44	3830,62
6	4085,31	2416,6	3975,13
8	-	3026,67	-
10	-	3135,45	-
12	-	3831,45	-
15	-	4000	-

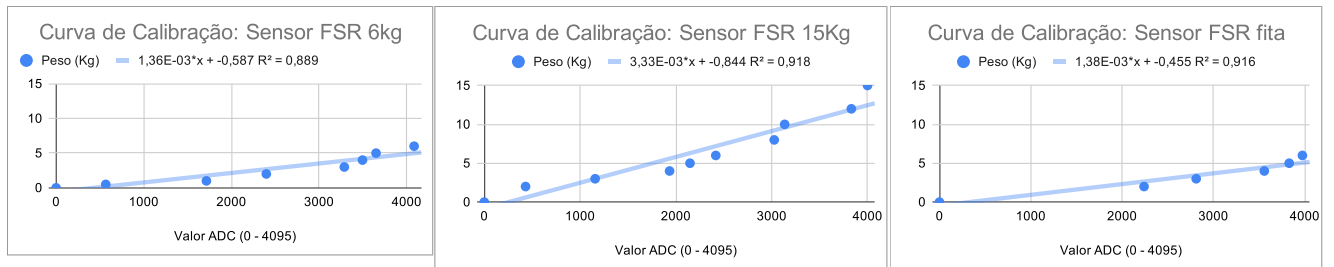


Figure 7: Correlation between ADC Counts and Force with Linear Fit

- **App:** It is possible to use the application to visualize mock data in three dimensions.

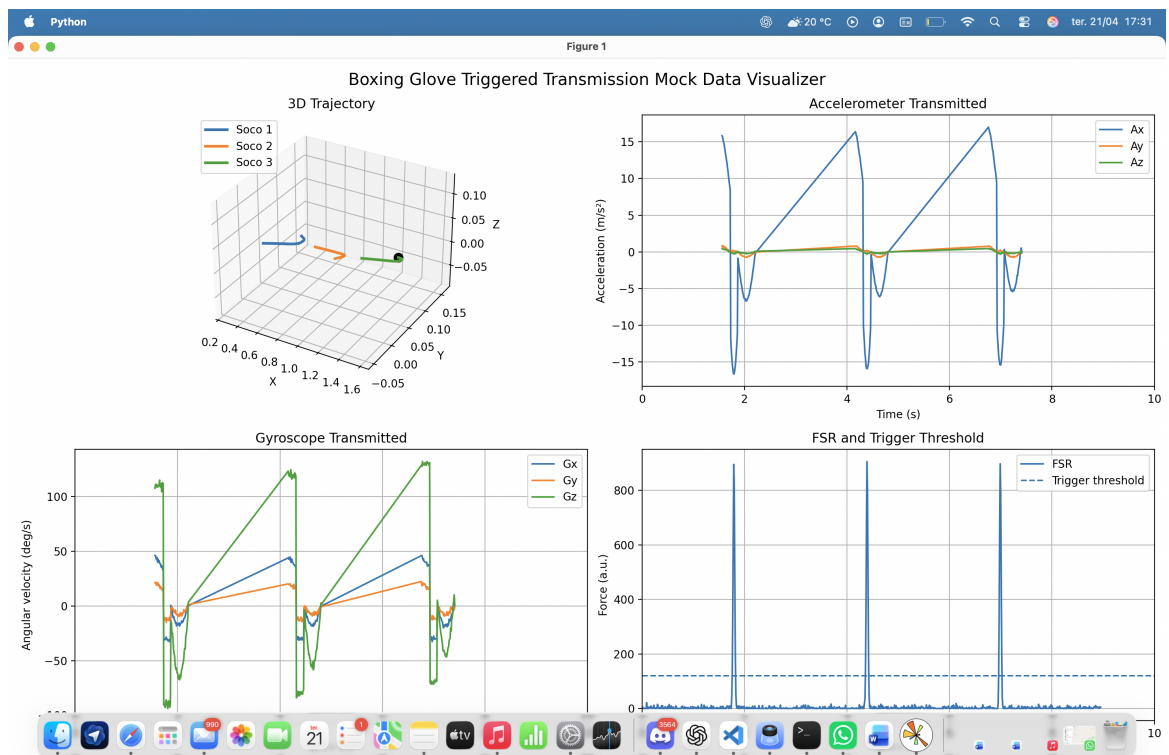


Figure 8: Software App Preview

7 Problem Identification and Technical Challenges

During the development phase, some technical obstacles were encountered that required specific debugging and analysis:

- **ITG3200 Logic Stagnation:** Despite a successful I2C handshake, confirmed by reading the correct device ID using the "Who Am I" register, the sensor output only returns zeros. Both the 3-axis gyroscope registers and the ITG3200 temperature sensor consistently return null values. We have adjusted the initialization sequence to the recommended 50 ms turn-on time and verified the power management configurations. Since even the temperature register is not updating, we are considering the possibility of a hardware malfunction in this specific unit.
- **ADXL345 Interrupt Persistence:** We encountered a critical issue when implementing hardware-controlled synchronization using the *DATA_READY* interrupt. Although the interrupt pin (A_INT1) was correctly measured at 3.3V (active high by default), it does not return to 0V after the host processor performs a burst read from the data register. Testing revealed that manually connecting the interrupt line to GND successfully triggers the sample processing task, but the sensor cannot independently generate the next transition. We are currently investigating if there is a conflict in the internal latching logic or a hardware issue with the pin's state transition.

8 Project Management and Progress

8.1 Work Distribution

To ensure efficiency, tasks were distributed based on each member's main area of expertise:

- **Rodrigo Figueira:** Hardware implementation, sensor characterization, power consumption analysis, strike timing study, and demos.
- **Tomás Rocha:** Firmware development (I2C drivers, BLE stack implementation) and Software development (Application, Error Handler and 3d Visualization)

8.2 Current Status and Planning

In this progress report, the project is aligned with the initial schedule. The hardware acquisition and basic firmware drivers (I2C, ADC, BLE) are completed. The application simulates sensor data through random generation and presents it in a three dimensional visualization environment.

The next phases will focus on the following:

- Integration of Gyroscope (ITG3200) data.
- Characterization and calibration of motion sensors.
- Final test with the FSR sensor to decide which type of sensor suits the best for our project.
- Integration of LED to indicate system status.
- Real-time trajectory reconstruction algorithms.
- Development of the application for data visualization.

9 Conclusion

The progress achieved so far confirms the technical feasibility of the smart boxing glove. The hardware architecture is established, and the system successfully captures data from the impact sensor, accelerometer, and magnetometer.

The coming weeks will be dedicated to resolving current inconsistencies with the ITG3200 gyroscope and improving hardware interrupt synchronization. Additionally, the project will advance to data transmission to the final visualization software and to completing the 3D trajectory reconstruction.

9.1 Source Code

All drivers and firmware logic developed for this project are hosted on GitHub. The repository can be accessed at: https://github.com/Th0masclassic/Box_Gloving_Sensor_Detection.

Bibliografia

- [1] Honeywell, “3-Axis Digital Compass IC HMC5883L,” Data Sheet, Form # 900405 Rev. E, Feb. 2013[cite: 3778, 3779]. [Online]. Disponível em: <https://www.magneticsensors.com>[cite: 3777].
- [2] Analog Devices, “ADXL345: 3-Axis, ± 2 g/ ± 4 g/ ± 8 g/ ± 16 g Digital Accelerometer,” Data Sheet, Rev. G[cite: 4, 61], May 2022. [Online]. Disponível em: <https://www.analog.com>.
- [3] InvenSense Inc., “ITG-3200 Product Specification,” Revision 1.3[cite: 1764], Mar. 2010. [Online]. Disponível em: <https://www.invensense.com>.
- [4] M. F. Sant, “GY-85: A Quick Datasheet Study”, *Medium: Jungletronics*, 2019. [Online]. Disponível em: <https://medium.com/jungletronics/gy-85-a-quick-datasheet-study-79019bb36fbf>.
- [5] Nologo Tech, “ESP32-C3 SuperMini Hardware Specification and Pinout”, 2024. [Online]. Disponível em: <https://www.nologo.tech/en/product/esp32/esp32c3SuperMini/>.
- [6] ESPBoards, “ESP32-C3 Super Mini Development Board Guide”, 2024. [Online]. Disponível em: <https://www.espboards.dev/esp32/esp32-c3-super-mini/>.
- [7] Espressif Systems, “NimBLE Stack API Reference”, *ESP-IDF Programming Guide*. [Online]. Disponível em: <https://docs.espressif.com/projects/esp-idf/en/stable/esp32/api-reference/bluetooth/nimble/index.html>.
- [8] D. Nedelkovski, “How to Track Orientation with ADXL345”, *HowToMechatronics*, 2021. [Online]. Disponível em: <https://howtomechatronics.com/tutorials/arduino/how-to-track-orientation-with-arduino-and-adxl345-accelerometer/>.
- [9] Control Everything Community, “ITG3200 Particle Library and Code Examples”, *GitHub Repository*. [Online]. Disponível em: <https://github.com/ControlEverythingCommunity/ITG3200/>.
- [10] Edaboard, “How to convert ADC value to weight/force”, *Electronic Design Forum*. [Online]. Disponível em: <https://www.edaboard.com/threads/how-to-convert-adc-value-to-weight.317018/>.
- [11] “Force Sensitive Resistor (FSR) - Interfacing and Code”, [Video]. YouTube. Disponível em: <https://www.youtube.com/watch?v=AA6RsMYHvk8>.